MRMhub Data Processing Workflow for Dataset 4

Steroid Assay

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1 Overview

Dataset 4 is a **fully quantitative steroid panel** used to validate MRMhub against a vendor pipeline (Agilent MassHunter). Fifteen steroids were measured in human serum with the IBL International *Steroid Panel LC-MS* kit (cat. 30191875) (IBL International GmbH 2022) according to the instructions for use, with modifications. The kit acquires its analytes over two chromatographic panels; **only Panel 1 (15 analytes) is used here**, which is why every sample identifier carries the `_P1` suffix (e.g. `Cal A_P1`, `QC_High1_P1`).

Unlike Datasets 1 and 3 (relative lipidomics), this is an externally calibrated absolute assay. The analytical design comprises an **external calibration curve** (`Cal 0`–`Cal F`), **low and high quality-control samples** (`QC_Low1`, `QC_High1`), and **external quality-assessment (EQA) samples** — SKML ring-trial materials with reference-method-assigned target concentrations (Jansen et al. 2014).

The workflow compares MRMhub and MassHunter at both pipeline stages:

- **INTEGRATOR** — peak areas from MRMhub-INTEGRATOR vs a MassHunter re-integration of the same raw files (Figure 7).
- **QUANT** — calibration fits and concentrations, ending in accuracy against the SKML EQA reference values (Figure 12).

2 Raw Data Processing: Peak Picking and Integration

Raw `.d` files were converted to mzML and integrated with MRMhub-INTEGRATOR, as described for Datasets 1 and 3. The batch was integrated in two subsets — the **system-suitability** injections (`SST`, ten replicate injections) and the **assay** samples (`ASSAY`: blanks, calibrators, QCs, EQA samples) — exported as separate long-format tables and combined downstream. INTEGRATOR reports peak areas, retention times and peak metadata in `long.csv`, and writes per-transition integration plots (the shaded, integrated region) to the `by_transition` folder.

The same raw files were independently re-integrated in Agilent MassHunter Quantitative Analysis, providing the reference peak areas and concentrations used throughout this document.

> **Note**
>
> The example chromatograms are cropped from the INTEGRATOR `by_transition` PDFs by `scripts/dataset4-chromatograms/dataset4-chromatograms.R` (run once; requires the `magick` R package and **ghostscript**) and embedded here as static images from `images/dataset4/`. The notebook render itself needs neither.

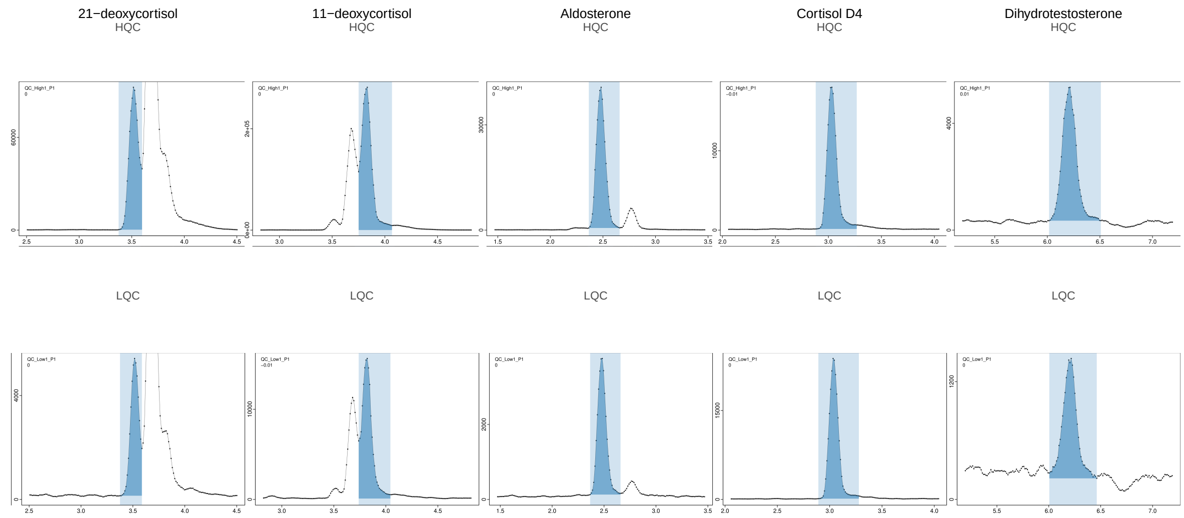

Figure 1: **Example INTEGRATOR chromatograms.** MRMhub-INTEGRATOR chromatograms for 21-deoxycortisol, 11-deoxycortisol, Aldosterone, Cortisol D4 (internal standard) and Dihydrotestosterone at high (HQC) and low (LQC) QC levels; the shaded region is the integrated peak.

3 Data Postprocessing and QC

Unlike Datasets 1 and 3, this is a short pipeline: a 15-analyte, single-batch panel needs no drift/batch correction or feature filtering. The QUANT module (`library(mrmhub)`) imports the INTEGRATOR results and metadata, normalizes to the internal standards, fits external calibration curves, and quantifies.

mrmhub version

This notebook was rendered with **mrmhub 0.9.8** (QUANT module of MRMhub). This API is currently on the `development` branch; install with `remotes::install_github("SLINGhub/MRMhub", ref = "development")`. The coloured console output additionally uses the `fansi` package (`install.packages("fansi")`).

```
library(mrmhub)
library(mirai) # parallel processing (picked up automatically by mrmhub)

# Colour mrmhub's cli console feedback in the rendered HTML (requires the fansi
↪ package)
# Colour the mrmhub console output in HTML; disable it for PDF, where the ANSI
# colour escapes would break LaTeX/lualatex compilation.
if (knitr::is_latex_output()) options(cli.num_colors = 1, crayon.enabled = FALSE)
↪ else mrmhub_enable_cli_color()
```

```r
# Set number of cores for parallel processing (all-but-one, so the system stays
↪  responsive)
n_cores <- {
  n <- parallel::detectCores()
  if (is.na(n)) n <- 4L
  max(1L, n - 1L)
}
if (mirai::status()$daemons == 0) mirai::daemons(n_cores)
```

3.1 Import INTEGRATOR results and metadata

The two INTEGRATOR subsets share columns and hold disjoint analyses, so they are combined by row-binding. The combined table is written to output/ because it is re-imported later (Figure 12) with MassHunter areas substituted in.

```r
sst_long <- read_csv(
  file.path(d4, "Dataset4_MRMhub-INTEGRATOR_SST.csv"),
  show_col_types = FALSE
)
assay_long <- read_csv(
  file.path(d4, "Dataset4_MRMhub-INTEGRATOR_ASSAY.csv"),
  show_col_types = FALSE
)
long_combined <- bind_rows(sst_long, assay_long)

long_combined_path <- file.path("output", "Dataset4_long_combined.csv")
write_csv(long_combined, long_combined_path)

# Analysis ids per subset (import strips the ".mzML" extension).
sst_analysis_ids <- unique(str_remove(
  sst_long$raw_data_filename,
  "[.]mzML$"
))
assay_analysis_ids <- unique(str_remove(
  assay_long$raw_data_filename,
  "[.]mzML$"
))

meta_path <- file.path(d4, "Dataset4_Metadata.xlsx")

mexp <- MRMhubExperiment(title = "Dataset 4 - Steroid Assay")
mexp <- import_data_mrmhub(
  mexp,
  path = long_combined_path,
  import_metadata = TRUE
```

```
)
#> v Imported 30 analyses with 64 features.
#> i feature_area selected as default feature intensity. Modify with
↪  `set_intensity_var()`.
#> v Analysis metadata associated with 30 analyses.
#> v Feature metadata associated with 64 features.
mexp <- import_metadata_msorganiser(
  mexp,
  path = meta_path,
  excl_unmatched_analyses = TRUE, ignore_warnings = TRUE
)
#> Found no errors, 2 warnings, and 2 notes in the metadata.
#> ----------------------------------------------------------------
#>   Type  Table    Column      Issue                         Count
#> 1 W*    Analyses analysis_id Analyses not in analysis data     2
#> 2 W*    Features feature_id  Feature(s) without metadata      34
#> 3 N     Analyses sample_id   Not defined for all analyses     16
#> 4 N     Features analyte_id  Not defined for all features     15
#>
#> ----------------------------------------------------------------
#> E = Error, W = Warning, W* = Suppressed Warning, N = Note
#> ----------------------------------------------------------------
#> v Analysis metadata associated with 30 analyses.
#> v Feature metadata associated with 30 features.
#> v Internal Standard metadata associated with 15 ISTDs.
#> v QC concentration metadata associated with 13 samples and 15 analytes
```

3.2 Normalization and calibration

Each feature is normalized to its stable-isotope-labelled internal standard, then a quadratic, `1/x`-weighted external calibration curve is fitted per analyte from the `Cal 0`–`Cal F` levels.

```
mexp <- normalize_by_istd(mexp)
#> v 15 features normalized with 15 ISTDs in 30 analyses.
mexp <- calc_calibration_results(
  mexp,
  fit_overwrite = TRUE,
  fit_model = "quadratic",
  fit_weighting = "1/x"
)
#> v Calibration curve fits calculated for all 15 quantifier features. Average r²:
↪  0.9978.
```

The fitted curves and their metrics are inspected before quantification.

```r
plot_calibrationcurves(
  mexp,
  fit_overwrite = FALSE,
  fit_model = "quadratic",
  fit_weighting = "1/x",
  include_istd = FALSE,
  include_qualifier = FALSE,
  show_progress = FALSE
)
#> v Calibration curve fits calculated for all 15 quantifier features. Average r²:
↪  0.9978.
#> v Done
```

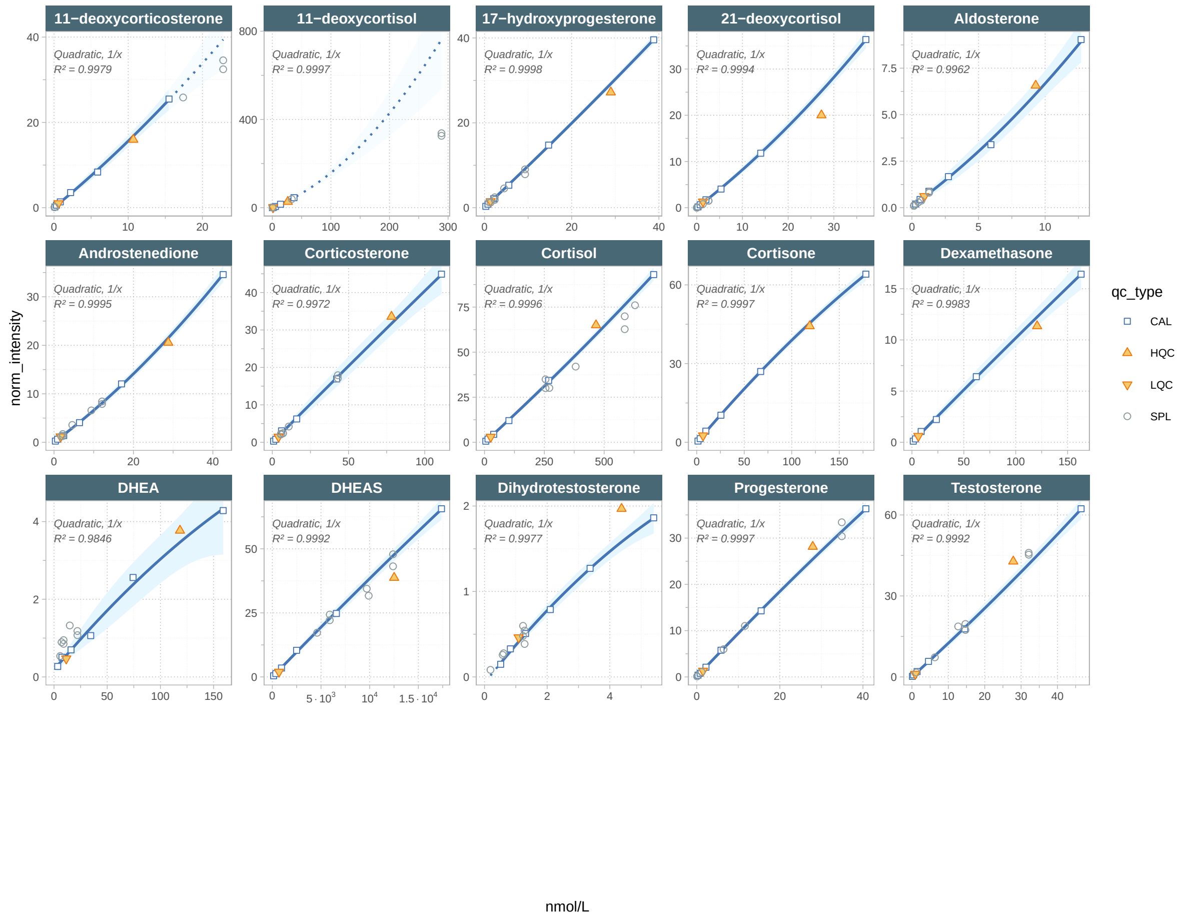

Figure 2: **External calibration curves.** Quadratic, 1/x-weighted curves for the 15 steroid analytes (quantifier transitions).

Table 1: **Calibration metrics per analyte.** Fit model, weighting, R^2, LOD and LOQ for each quantifier.

```
get_calibration_metrics(mexp, include_qualifier = FALSE, summary_table = TRUE)
#> # A tibble: 15 x 6
#>    analyte                fit_model fit_weighting    r2   lod    loq
#>    <chr>                  <chr>     <chr>         <dbl> <dbl>  <dbl>
#>  1 11-deoxycorticosterone quadratic 1/x           0.998 0.457  1.38
#>  2 11-deoxycortisol       quadratic 1/x           1.000 0.279  0.844
#>  3 17-hydroxyprogesterone quadratic 1/x           1.000 0.197  0.596
#>  4 21-deoxycortisol       quadratic 1/x           0.999 0.427  1.29
#>  5 Aldosterone            quadratic 1/x           0.996 0.639  1.94
#>  6 Androstenedione        quadratic 1/x           0.999 0.445  1.35
#>  7 Corticosterone         quadratic 1/x           0.997 1.25   3.79
#>  8 Cortisol               quadratic 1/x           1.000 1.22   3.71
#>  9 Cortisone              quadratic 1/x           1.000 0.463  1.4
#> 10 DHEA                   quadratic 1/x           0.985 3.03   9.18
#> 11 DHEAS                  quadratic 1/x           0.999 8.29  25.1
#> 12 Dexamethasone          quadratic 1/x           0.998 1.19   3.6
#> 13 Dihydrotestosterone    quadratic 1/x           0.998 0.178  0.54
#> 14 Progesterone           quadratic 1/x           1.000 0.251  0.76
#> 15 Testosterone           quadratic 1/x           0.999 0.458  1.39
```

3.3 Quantification

Concentrations are computed from the normalized intensities via the calibration curves. Qualifier transitions are excluded; analytes whose curve failed to fit are skipped rather than aborting the run.

```
mexp <- quantify_by_calibration(
  mexp,
  fit_overwrite = FALSE,
  include_qualifier = FALSE,
  ignore_failed_calibration = TRUE,
  fit_model = "quadratic",
  fit_weighting = "1/x"
)
#> v Calibration curve fits calculated for all 15 quantifier features. Average r²:
↪  0.9978.
#> i 77 concentration values fall outside the calibrated range (retained, flagged in
↪  feature_conc_out_of_range).
#> v Concentrations calculated for 15 features in 30 analyses.
#> v Concentrations are given in nmol/L.

# Export + reload the concentration table (wide: analysis_id, qc_type, analytes).
conc_path <- file.path("output", "Dataset4_quant_results.csv")
save_dataset_csv(
```

```
  mexp,
  path = conc_path,
  variable = "conc",
  add_qctype = TRUE
)
#> v Concentration values for 30 analyses and 15 features have been exported to
 ↪  'output/Dataset4_quant_results.csv'.
d_conc_raw <- read_csv(conc_path, show_col_types = FALSE)
```

3.4 MassHunter re-integration

The MassHunter Quantitative Analysis export is imported as a second experiment and reshaped to one tidy table — the single MassHunter source for all comparisons below. The trailing `(ISTD)` suffix on labelled internal standards is stripped so feature identifiers match MRMhub.

```
mh_reint <- import_data_masshunter(
  MRMhubExperiment(title = "MassHunter re-integration"),
  path = file.path(
    d4,
    "Dataset4_IntegrationResults_MassHunter_IsoSP-22_V3.csv"
  ),
  import_metadata = TRUE, silent = TRUE
)
#> i feature_area selected as default feature intensity. Modify with
 ↪  `set_intensity_var()`.
#> v Analysis metadata associated with 30 analyses.
#> v Feature metadata associated with 30 features.
mh_data <- get_analyticaldata(mh_reint, annotated = FALSE) |>
  transmute(
    analysis_id,
    feature_id = str_remove(feature_id, "[ ]*\\(ISTD\\)$"),
    rt_mh = feature_rt, area_mh = feature_area,
    conc_mh = feature_conc_final, mi_mh = feature_manual_integration
  )
```

3.5 Below-Cal-A exclusion for study samples

Study samples (`SPL` = the SKML EQA samples) whose **quantifier peak area falls below the corresponding `Cal A` area** are below the lowest non-zero calibrator and treated as not reliably quantifiable. The rule is applied per analyte and **independently per method** — MRMhub SPL area vs MRMhub Cal A area, MassHunter SPL area vs MassHunter Cal A area — yielding the flags `excl_mrm` / `excl_mh` that downstream comparisons apply to their SPL rows. Internal standards are never filtered.

```r
CAL_A_ID <- "Cal A_P1"

# MRMhub quantifier areas (quantifiers only, excluding the ISTDs).
mrm_q_area <- get_analyticaldata(mexp, annotated = TRUE) |>
  filter(is_quantifier, !is_istd) |>
  transmute(
    analysis_id,
    analyte_id = feature_id, qc_type, area_mrm = feature_area
  )

cala_mrm <- mrm_q_area |>
  filter(analysis_id == CAL_A_ID) |>
  select(analyte_id, cala_area_mrm = area_mrm)
cala_mh <- mh_data |>
  filter(analysis_id == CAL_A_ID) |>
  transmute(analyte_id = feature_id, cala_area_mh = area_mh)

# One row per SPL (analysis, analyte); flag below-Cal-A per method (NA-safe).
spl_area_excl <- mrm_q_area |>
  filter(qc_type == "SPL") |>
  left_join(
    select(mh_data, analysis_id, analyte_id = feature_id, area_mh),
    by = c("analysis_id", "analyte_id")
  ) |>
  left_join(cala_mrm, by = "analyte_id") |>
  left_join(cala_mh, by = "analyte_id") |>
  mutate(
    excl_mrm = !is.na(area_mrm) & !is.na(cala_area_mrm) &
      area_mrm < cala_area_mrm,
    excl_mh = !is.na(area_mh) & !is.na(cala_area_mh) &
      area_mh < cala_area_mh
  ) |>
  select(
    analysis_id, analyte_id,
    area_mrm, area_mh, cala_area_mrm, cala_area_mh,
    excl_mrm, excl_mh
  )
```

3.6 Low / high QC

Quantified concentrations of the low and high QC samples, one row per analyte.

Table 2: **Low/high QC concentrations.** Quantified concentrations of the low and high QC samples, one row per analyte.

```
d_conc_raw |>
  filter(qc_type %in% c("LQC", "HQC")) |>
  pivot_longer(
    !c(analysis_id, qc_type),
    names_to = "analyte", values_to = "conc"
  ) |>
  mutate(qc = factor(qc_type, levels = c("LQC", "HQC"))) |>
  select(analyte, qc, conc) |>
  pivot_wider(names_from = qc, values_from = conc) |>
  mutate(across(where(is.numeric), ~ signif(.x, 3)))
#> # A tibble: 15 x 3
#>    analyte                    LQC      HQC
#>    <chr>                    <dbl>    <dbl>
#>  1 11-deoxycorticosterone   0.672    10.2
#>  2 11-deoxycortisol         1.45     23.5
#>  3 17-hydroxyprogesterone   1.51     27.1
#>  4 21-deoxycortisol         1.68     22.4
#>  5 Aldosterone              0.988     9.86
#>  6 Androstenedione          1.84     27.7
#>  7 Corticosterone           3.6      82.8
#>  8 Cortisol                24.9     505
#>  9 Cortisone                6.27    116
#> 10 Dexamethasone            5.98    112
#> 11 DHEA                     8.94    132
#> 12 DHEAS                  521     10100
#> 13 Dihydrotestosterone      1.21      5.86
#> 14 Progesterone             1.32     31.1
#> 15 Testosterone             0.856    32.8
```

4 MRMhub vs MassHunter — Peak Integration

Figure 7 compares MRMhub-INTEGRATOR peak areas with the MassHunter re-integration of the same raw files. MRMhub quantifier and ISTD peaks are joined to MassHunter on `analysis_id` + `feature_id`, dropping near-noise peaks (area < 100), empty CAL rows, and below-Cal-A SPL points.

```
# Manuscript house theme + palettes (shared by both figures).
theme_set(
  theme_bw(base_size = 8, base_family = "sans") +
    theme(
      panel.grid.minor = element_blank(),
      legend.position = "bottom",
      plot.title = element_text(size = 8, face = "plain"),
      legend.key.size = unit(3.2, "mm"), legend.margin = margin(1, 1, 1, 1)
    )
)

col_method <- c(MRMhub = "#2C7FB8", MassHunter = "#D95F0E")
smp_cols <- c(
  CAL = "#0072B2",
  HQC = "#D55E00", LQC = "#E69F00", SPL = "grey30", SST = "#009E73",
  SBLK = "#CC79A7", PBLK = "#999999"
)

# Save a composite as vector PDF + 300-dpi PNG at a given size (mm) for the
↪ manuscript.
save_fig <- function(plot, stem, height, width = 180) {
  ggsave(
    file.path("output", paste0(stem, ".pdf")),
    plot,
    device = cairo_pdf, width = width, height = height, units = "mm"
  )
  ggsave(
    file.path("output", paste0(stem, ".png")),
    plot,
    device = ragg::agg_png, width = width, height = height,
    units = "mm", dpi = 300
  )
  invisible(plot)
}
```

```
mrm_peaks_q <- get_analyticaldata(mexp, annotated = TRUE) |>
  filter(is_quantifier | is_istd) |>
  transmute(
    analysis_id,
    feature_id, qc_type, sample_id,
```

```
    feature_type = factor(
      if_else(is_istd, "ISTD", "Quantifier"),
      levels = c("Quantifier", "ISTD")
    ),
    rt_mrm = feature_rt, area_mrm = feature_area
  )

# Peak-level MRMhub MassHunter comparison table, built up by a filter cascade so
# only well-integrated, comparable peaks survive.
d_peakcmp_reint <- mrm_peaks_q |>
  # Keep only peaks both integrators reported for the same analysis + feature.
  inner_join(mh_data, by = c("analysis_id", "feature_id")) |>
  filter(
    !is.na(rt_mrm),
    !is.na(rt_mh), !is.na(area_mrm), !is.na(area_mh)
  ) |>
  # Drop near-noise peaks (either method below area 100).
  filter(area_mrm >= 100, area_mh >= 100) |>
  # Drop empty (unassigned) calibrator rows; ISTDs are always kept.
  filter(!(qc_type == "CAL" & is.na(sample_id) & feature_type != "ISTD")) |>
  left_join(
    select(spl_area_excl, analysis_id, analyte_id, excl_mrm, excl_mh),
    by = c("analysis_id", "feature_id" = "analyte_id")
  ) |>
  # Drop below-Cal-A (below-LOQ) study samples flagged by either method.
  filter(
    !(qc_type == "SPL" &
      (coalesce(excl_mrm, FALSE) | coalesce(excl_mh, FALSE)))
  ) |>
  select(-excl_mrm, -excl_mh) |>
  mutate(
    area_diff_pct = (area_mh - area_mrm) / area_mrm * 100,
    rt_diff = rt_mh - rt_mrm,
    # Tag each analysis as an SST (system-suitability) or ASSAY run.
    subset = case_when(
      analysis_id %in% sst_analysis_ids ~ "SST",
      analysis_id %in% assay_analysis_ids ~ "ASSAY", TRUE ~ NA_character_
    )
  )
```

4.1 Peak-area agreement

Log–log peak areas against the 1:1 line (Pearson r on log10 values). CAL/QC/SST samples span four orders of magnitude, and the two integrators agree closely.

```r
smp_lv <- c("CAL", "HQC", "LQC", "SST")
smp_cols4 <- c(
  CAL = "#0072B2",
  HQC = "#D55E00", LQC = "#E69F00", SST = "#009E73"
)
ft_shapes <- c(Quantifier = 16, ISTD = 17)
scale_smp <- scale_colour_manual(
  values = smp_cols4,
  limits = smp_lv, drop = FALSE
)
scale_ft <- scale_shape_manual(values = ft_shapes)

d_cmp <- d_peakcmp_reint |>
  filter(qc_type %in% smp_lv) |>
  mutate(qc_type = factor(qc_type, levels = smp_lv))
r_log <- cor(log10(d_cmp$area_mrm), log10(d_cmp$area_mh))
ax_lims <- range(c(d_cmp$area_mrm, d_cmp$area_mh))
p_corr <- d_cmp |>
  ggplot(aes(area_mrm, area_mh)) +
  geom_abline(
    slope = 1,
    intercept = 0, linetype = "dashed", colour = "grey40"
  ) +
  geom_point(
    aes(colour = qc_type, shape = feature_type),
    size = 1.1, alpha = 0.7
  ) +
  # Ring the peaks MassHunter integrated manually (MI flag), keeping the
  # sample-type colour / feature shape encoding underneath.
  geom_point(
    data = ~ filter(.x, mi_mh),
    aes(alpha = "Manual integration (MassHunter)"), shape = 1,
    colour = "grey15", size = 2.3, stroke = 0.5
  ) +
  annotate(
    "text",
    x = ax_lims[1], y = ax_lims[2], hjust = 0, vjust = 1, size = 2.4,
    label = sprintf("Pearson r = %.4f", r_log)
  ) +
  scale_x_log10(limits = ax_lims) +
  scale_y_log10(limits = ax_lims) +
  scale_smp +
  scale_ft +
  scale_alpha_manual(
    name = NULL,
    values = c("Manual integration (MassHunter)" = 1)
  ) +
  guides(
```

```
    colour = guide_legend(order = 1, nrow = 2),
    shape = guide_legend(order = 2, nrow = 2),
    alpha = guide_legend(
      order = 3,
      override.aes = list(
        shape = 1,
        colour = "grey15", size = 2.3, stroke = 0.5
      )
    )
  ) +
  labs(
    x = "MRMhub peak area",
    y = "MassHunter peak area", colour = "Sample type", shape = "Feature",
    title = "Peak-area agreement"
  )
p_corr
```

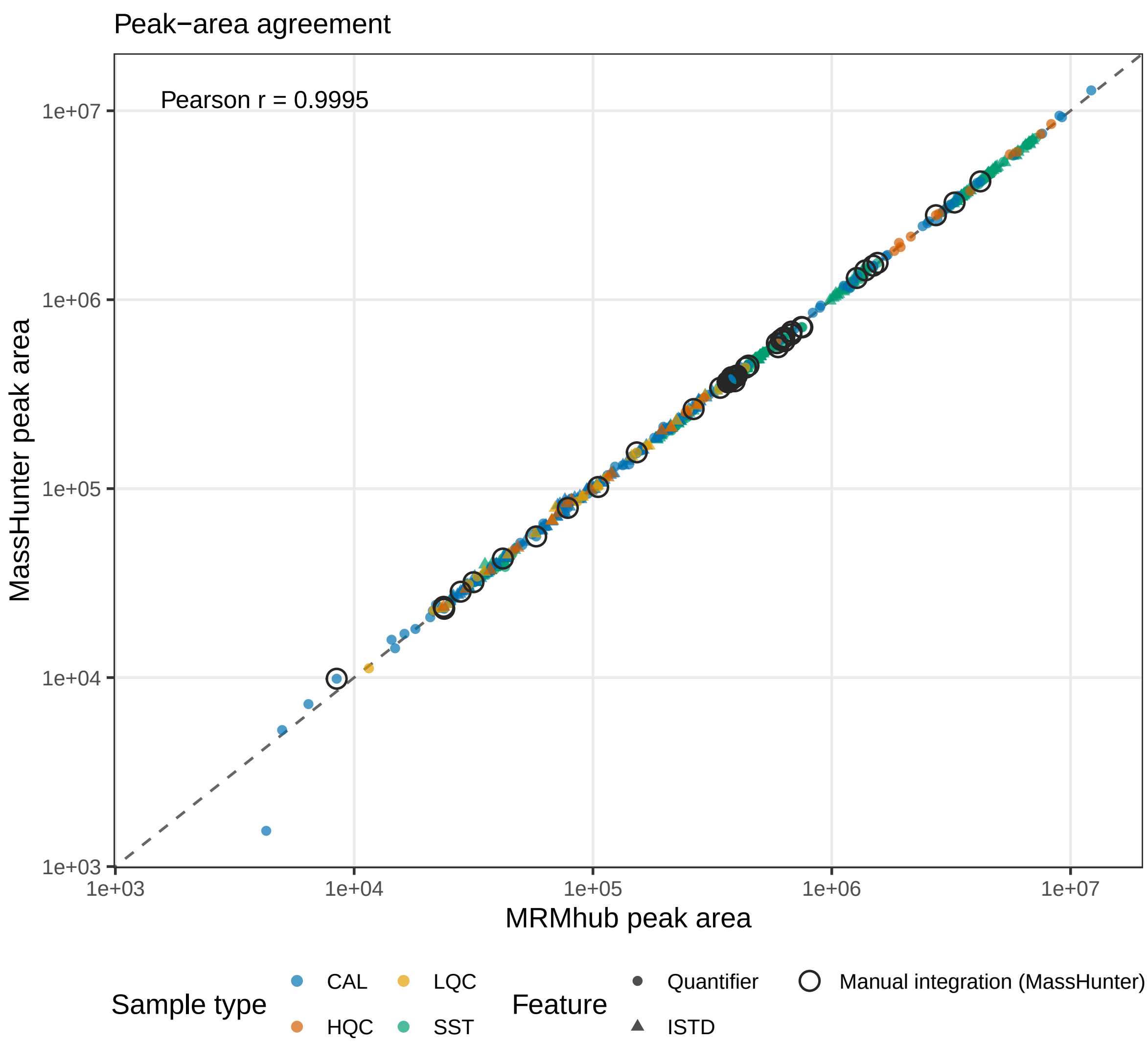

Figure 3: **Peak-area agreement.** MRMhub vs MassHunter peak areas (log–log) against the 1:1 line; Pearson *r* on log10 values across CAL/QC/SST samples. Open grey rings mark peaks MassHunter integrated manually. Panel A of Figure 7.

4.2 Precision (%CV) equivalence

Per-feature %CV over the ten replicate SST injections, MRMhub vs MassHunter. The two methods show comparable precision, with points scattering around the 1:1 line.

```
sst_cv <- d_peakcmp_reint |>
  filter(subset == "SST") |>
  group_by(feature_type, feature_id) |>
  summarise(
```

```
    cv_mrmhub = sd(area_mrm) / mean(area_mrm) * 100,
    cv_masshunter = sd(area_mh) / mean(area_mh) * 100, .groups = "drop"
  )
cv_lims <- range(c(sst_cv$cv_mrmhub, sst_cv$cv_masshunter)) + c(-0.3, 0.3)
p_cv <- sst_cv |>
  ggplot(aes(cv_mrmhub, cv_masshunter, shape = feature_type)) +
  geom_abline(
    slope = 1,
    intercept = 0, linetype = "dashed", colour = "grey60"
  ) +
  geom_point(size = 1.6, alpha = 0.85, colour = "#009E73") +
  coord_cartesian(xlim = cv_lims, ylim = cv_lims) +
  scale_ft +
  guides(shape = "none") +
  labs(
    x = "MRMhub %CV (10 SST)",
    y = "MassHunter %CV", title = "Precision (%CV) equivalence"
  )
p_cv
```

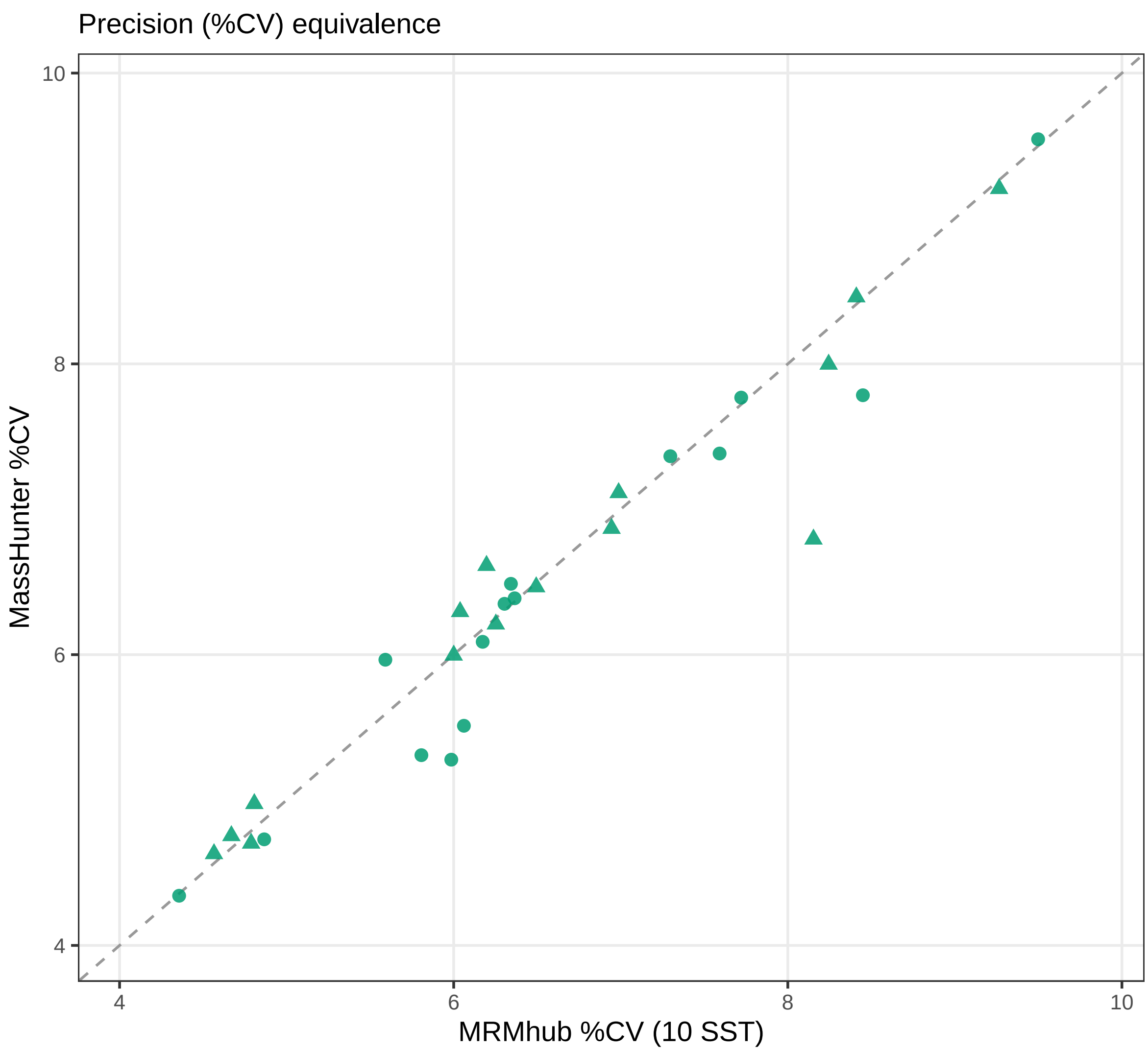

Figure 4: **Precision (%CV) equivalence.** Per-feature %CV over the ten replicate SST injections, MRMhub vs MassHunter; points on the 1:1 line indicate equal precision. Panel B of Figure 7.

4.3 Bland–Altman

Percent difference (MassHunter − MRMhub) against the mean area, excluding SST and near-noise peaks (mean area $< 10^3$). The three highlighted analytes are among those with the largest real-sample differences (see Table 3); their chromatograms are shown below.

```
highlight_analytes <- c(
  "21-deoxycortisol",
  "11-deoxycortisol", "Aldosterone"
```

```
)
hl_cols <- setNames(c("#D81B60", "#1E88E5", "#8E44AD"), highlight_analytes)

d_ba <- d_peakcmp_reint |>
  filter(qc_type != "SST") |>
  mutate(
    mean_area = (area_mrm + area_mh) / 2,
    pct_diff = (area_mh - area_mrm) / mean_area * 100,
    hl = factor(
      if_else(
        feature_id %in% highlight_analytes,
        feature_id, NA_character_
      ),
      levels = highlight_analytes
    )
  ) |>
  filter(mean_area >= 1e3)
ba_bias <- mean(d_ba$pct_diff)
ba_loa <- ba_bias + c(-1.96, 1.96) * sd(d_ba$pct_diff)
ba_xr <- max(d_ba$mean_area)
p_ba <- d_ba |>
  ggplot(aes(mean_area, pct_diff)) +
  geom_hline(yintercept = 0, colour = "grey70") +
  geom_hline(yintercept = ba_bias, colour = "grey20") +
  geom_hline(yintercept = ba_loa, linetype = "dashed", colour = "grey50") +
  geom_point(
    data = ~ filter(.x, is.na(hl)),
    colour = "grey80", size = 0.9, alpha = 0.6
  ) +
  geom_point(
    data = ~ filter(.x, !is.na(hl)),
    aes(colour = hl), size = 1.7, alpha = 0.9
  ) +
  # Ring the peaks MassHunter integrated manually (MI flag).
  geom_point(
    data = ~ filter(.x, mi_mh),
    aes(alpha = "Manual integration (MassHunter)"), shape = 1,
    colour = "grey15", size = 2.0, stroke = 0.5
  ) +
  annotate(
    "text",
    x = ba_xr, y = ba_bias, hjust = 1, vjust = -0.6, size = 2.1,
    colour = "grey20", label = sprintf("mean bias %.1f%%", ba_bias)
  ) +
  annotate(
    "text",
    x = ba_xr, y = ba_loa[2], hjust = 1, vjust = 1.4, size = 2,
    colour = "grey45", label = "+95% LoA"
```

```
  ) +
  annotate(
    "text",
    x = ba_xr, y = ba_loa[1], hjust = 1, vjust = -0.7, size = 2,
    colour = "grey45", label = "-95% LoA"
  ) +
  scale_colour_manual(
    values = hl_cols,
    name = "Highlighted analyte", na.translate = FALSE
  ) +
  scale_alpha_manual(
    name = NULL,
    values = c("Manual integration (MassHunter)" = 1)
  ) +
  scale_x_log10(limits = c(1e3, NA)) +
  coord_cartesian(ylim = c(-20, 20)) +
  guides(
    alpha = guide_legend(
      order = 2,
      override.aes = list(
        shape = 1,
        colour = "grey15", size = 2.0, stroke = 0.5
      )
    )
  ) +
  labs(
    x = "Mean peak area of the two methods (>= 10^3)",
    y = "Percent difference (MH - MRMhub) %",
    title = sprintf(
      "Bland-Altman (excl. SST, area >= 10^3): mean bias %.1f%%, 95%% LoA [%.0f,
       ↪  %.0f]%%",
      ba_bias, ba_loa[1], ba_loa[2]
    )
  )
p_ba
```

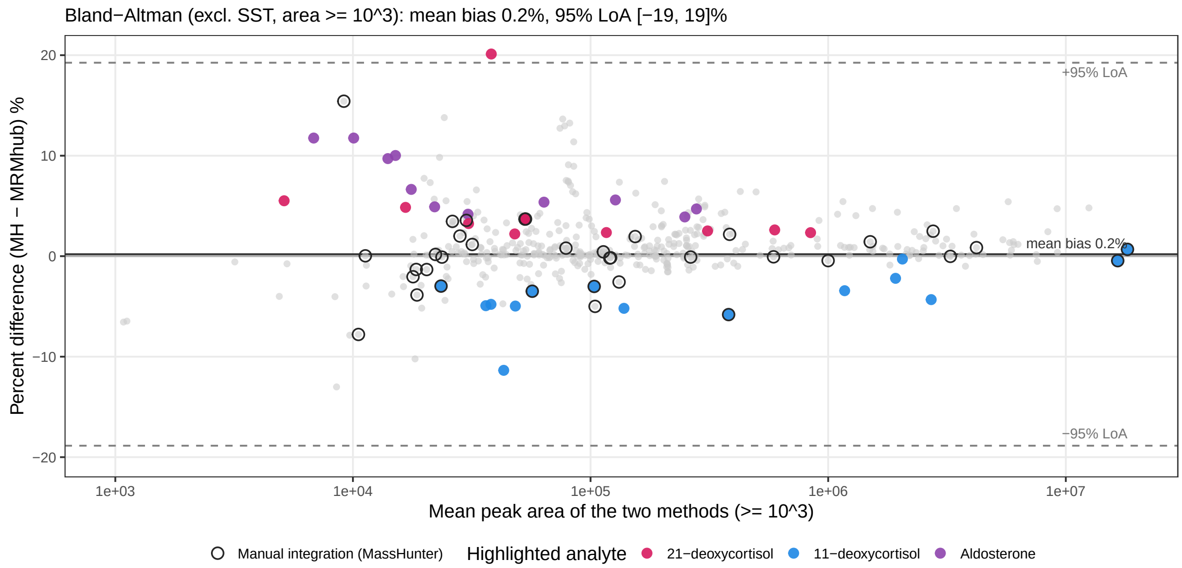

Figure 5: **Bland–Altman of real-sample differences.** Percent difference (MassHunter − MRMhub) against the mean peak area (excl. SST; mean area 10^3); the three largest-difference analytes are highlighted and open grey rings mark manual MassHunter integrations. Panel C of Figure 7.

4.4 Example chromatograms

The largest real-sample area differences occur where a peak sits on an elevated baseline or beside a co-eluting neighbour, so the integration boundaries — and hence the areas — legitimately differ between the two integrators. The candidate table ranks these cases; the panel crops the three chosen examples (one per highlighted analyte) from the MRMhub `by_transition` chromatograms.

```r
# Panel-D chromatograms are the pre-generated crops (see the INTEGRATOR section);
# subtitles carry the data-derived area difference for each example.
chrom_examples <- tibble::tribble(
  ~name,
  ~feature_id, ~analysis_id, "fig1d_21-deoxycortisol", "21-deoxycortisol",
  "SKML2025_6B_P1", "fig1d_11-deoxycortisol", "11-deoxycortisol",
  "SKML2026_2B_P1", "fig1d_Aldosterone", "Aldosterone", "SKML2026_2A_P1"
)

chrom_meta <- chrom_examples |>
  left_join(
    distinct(d_peakcmp_reint, feature_id, analysis_id, area_diff_pct),
    by = c("feature_id", "analysis_id")
  ) |>
  mutate(
    subtitle = sprintf(
      "%s  (dArea %+.1f%%)",
```

Table 3: **Largest area-difference candidates.** Peaks ranked by absolute MRMhub–MassHunter area difference, from which the example chromatograms are chosen.

```r
chrom_candidates <- d_peakcmp_reint |>
  filter(
    feature_type == "Quantifier",
    qc_type %in% c("CAL", "HQC", "LQC", "SPL", "SST"), area_mrm >= 20000
  ) |>
  arrange(desc(abs(area_diff_pct))) |>
  transmute(
    feature_id,
    analysis_id, qc_type,
    area_mrm = round(area_mrm),
    area_mh = round(area_mh), area_diff_pct = round(area_diff_pct, 1)
  ) |>
  head(10)
chrom_candidates
#> # A tibble: 10 x 6
#>    feature_id             analysis_id     qc_type area_mrm area_mh area_diff_pct
#>    <chr>                  <chr>           <chr>      <dbl>   <dbl>         <dbl>
#>  1 21-deoxycortisol       SKML2025_6B_P1  SPL        34372   42059          22.4
#>  2 17-hydroxyprogesterone SKML2025_6B_P1  SPL        76446   59981         -21.5
#>  3 11-deoxycortisol       SKML2026_2B_P1  SPL        45558   40661         -10.7
#>  4 Dihydrotestosterone    Cal D_P1        CAL        22013   24289          10.3
#>  5 DHEA                   20260212_SST03  SST        42897   38529         -10.2
#>  6 DHEA                   Cal D_P1        CAL        81165   88766           9.4
#>  7 11-deoxycortisol       20260212_SST07  SST      1285343 1389585           8.1
#>  8 Androstenedione        Cal A_P1        CAL        76008   81960           7.8
#>  9 DHEA                   Cal E_P1        CAL       197230  212468           7.7
#> 10 17-hydroxyprogesterone SKML2026_2Ab_P1 SPL       127355  137090           7.6
```

```r
      str_remove(analysis_id, "_P1$"), area_diff_pct
    )
  )
chrom_plots <- Map(
  chrom_gg,
  chrom_meta$name, chrom_meta$feature_id, chrom_meta$subtitle
)
p_chrom <- wrap_elements(full = wrap_plots(chrom_plots, nrow = 1))
p_chrom
```

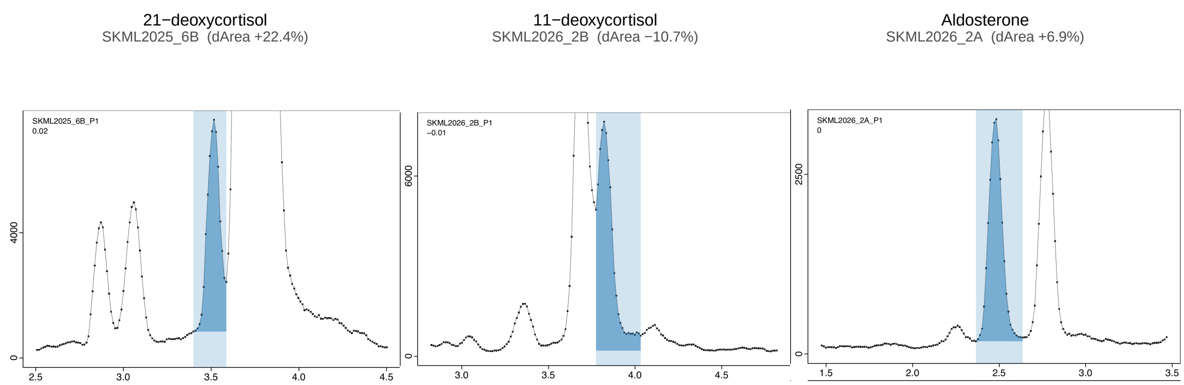

Figure 6: **Example chromatograms.** MRMhub chromatograms for the three highlighted analytes, where the integration boundaries — and hence the areas — legitimately differ between the two integrators. Panel D of Figure 7.

4.5 Figure 1 — manuscript figure

The four panels above are composed into the publication figure (**Figure 1**) and saved as a vector PDF and a 300-dpi PNG (180 mm double-column width).

```r
row_ab <- (p_corr + p_cv) +
  plot_layout(guides = "collect") &
  theme(legend.position = "bottom")
fig1 <- row_ab /
  p_ba /
  p_chrom +
  plot_layout(heights = c(1, 0.72, 0.58)) +
  plot_annotation(tag_levels = "A")

save_fig(fig1, "fig1_integration_correspondence", height = 230)
fig1
```

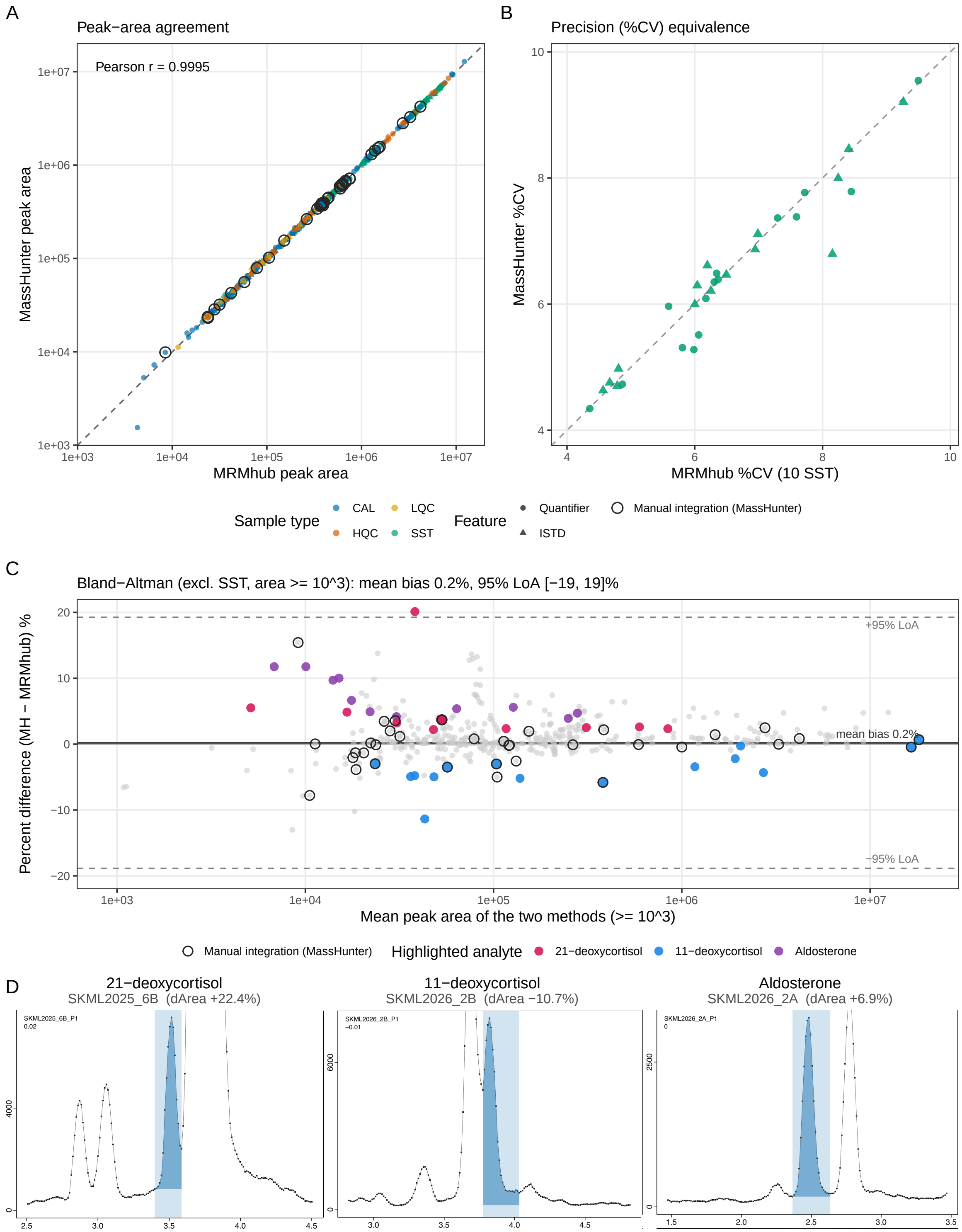

Figure 7: **MRMhub vs MassHunter — peak integration.** Composite of the four panels above; reproduced as the manuscript's **Figure 1**. **(A)** Peak-area agreement, **(B)** precision (%CV) equivalence, **(C)** Bland–Altman of real-sample differences, **(D)** example chromatograms. Exported to `output/fig1_integration_correspondence.{pdf,png}` (180 mm).

5 MRMhub vs MassHunter — Quantification

Figure 12 compares the two pipelines at the quantification stage. To isolate the post-processing (calibration + quantification) from the peak integration, we also feed **MassHunter's own peak areas through the MRMhub pipeline** — identical areas, identical calibration model — so any concentration difference is due to post-processing alone.

```r
# Rebuild the INTEGRATOR long table with MassHunter's peak areas swapped in for
# MRMhub's (matched on analysis + feature). Running the identical pipeline on this
# table - same areas, same calibration model - makes any concentration difference
# vs MassHunter attributable to post-processing alone, not to peak integration.
mh_area <- mh_data |>
  transmute(analysis_id, feature_name = feature_id, area_mh)
long_mh <- read_csv(long_combined_path, show_col_types = FALSE) |>
  mutate(.aid = str_remove(raw_data_filename, "[.]mzML$")) |>
  inner_join(mh_area, by = c(".aid" = "analysis_id", "feature_name")) |>
  mutate(area = area_mh) |>
  select(-.aid, -area_mh)

# The pipeline reads from disk, so write the area-substituted table to a temp file.
tmp_long <- tempfile(fileext = ".csv")
write_csv(long_mh, tmp_long)

mexp_mh <- MRMhubExperiment(title = "MassHunter areas via MRMhub pipeline")
mexp_mh <- import_data_mrmhub(
  mexp_mh,
  path = tmp_long,
  import_metadata = TRUE
)
#> v Imported 30 analyses with 30 features.
#> i feature_area selected as default feature intensity. Modify with
↪  `set_intensity_var()`.
#> v Analysis metadata associated with 30 analyses.
#> v Feature metadata associated with 30 features.
mexp_mh <- import_metadata_msorganiser(
  mexp_mh,
  path = meta_path,
  excl_unmatched_analyses = TRUE, ignore_warnings = TRUE
)
#> Found no errors, 1 warning, and 2 notes in the metadata.
#> ----------------------------------------------------------------
#>   Type  Table    Column      Issue                         Count
#> 1 W*    Analyses analysis_id Analyses not in analysis data     2
#> 2 N     Analyses sample_id   Not defined for all analyses     16
#> 3 N     Features analyte_id  Not defined for all features     15
#>
#> ----------------------------------------------------------------
#> E = Error, W = Warning, W* = Suppressed Warning, N = Note
```

```
#> ---------------------------------------------------------------
#> v Analysis metadata associated with 30 analyses.
#> v Feature metadata associated with 30 features.
#> v Internal Standard metadata associated with 15 ISTDs.
#> v QC concentration metadata associated with 13 samples and 15 analytes
mexp_mh <- normalize_by_istd(mexp_mh)
#> Warning: ! 2 feature values could not be ISTD-normalized because the internal
↪ standard
#>   had zero intensity and were set to NA.
#> i Affected analyses: "Blank2_P1" and "Blank3_P1"
#> v 15 features normalized with 15 ISTDs in 30 analyses.
mexp_mh <- calc_calibration_results(
  mexp_mh,
  fit_overwrite = TRUE,
  fit_model = "quadratic",
  fit_weighting = "1/x"
)
#> v Calibration curve fits calculated for all 15 quantifier features. Average r²:
↪ 0.9963.
mexp_mh <- quantify_by_calibration(
  mexp_mh,
  fit_overwrite = FALSE,
  include_qualifier = FALSE,
  ignore_failed_calibration = TRUE,
  fit_model = "quadratic",
  fit_weighting = "1/x"
)
#> v Calibration curve fits calculated for all 15 quantifier features. Average r²:
↪ 0.9963.
#> i 80 concentration values fall outside the calibrated range (retained, flagged in
↪ feature_conc_out_of_range).
#> v Concentrations calculated for 15 features in 30 analyses.
#> v Concentrations are given in nmol/L.
```

5.1 Calibration fit comparison

Calibration R^2 per analyte: MassHunter's own fit vs the MRMhub full-pipeline fit.

```
mh_own_r2 <- read_csv(
  file.path(d4, "Dataset4_MassHunter-CalibrationFits.csv"),
  show_col_types = FALSE
) |>
  transmute(analyte_id, r2_mh_own = r2)
r2_cmp <- get_calibration_metrics(mexp_mh, include_qualifier = FALSE) |>
  transmute(analyte_id = feature_id, r2_mrmpipe_mharea = r2) |>
  left_join(mh_own_r2, by = "analyte_id") |>
```

```
  left_join(
    get_calibration_metrics(mexp, include_qualifier = FALSE) |>
      transmute(analyte_id = feature_id, r2_mrmpipe_mrmarea = r2),
    by = "analyte_id"
  )

chain_lv <- c("MassHunter", "MRMhub")
chain_col <- c(MassHunter = "#D95F0E", MRMhub = "#2C7FB8")
ord_r2 <- r2_cmp |>
  arrange(r2_mrmpipe_mrmarea) |>
  pull(analyte_id)

d_r2_long <- r2_cmp |>
  transmute(
    analyte_id,
    MassHunter = r2_mh_own, MRMhub = r2_mrmpipe_mrmarea
  ) |>
  pivot_longer(-analyte_id, names_to = "chain", values_to = "r2") |>
  mutate(
    analyte_id = factor(analyte_id, levels = ord_r2),
    chain = factor(chain, levels = chain_lv)
  )

# Truncate the x-axis just below the dense 0.996-1.000 cluster; flag off-scale rows.
R2_FLOOR <- 0.9955
r2_off <- d_r2_long |>
  filter(r2 < R2_FLOOR) |>
  distinct(analyte_id) |>
  mutate(r2 = R2_FLOOR)
off_txt <- d_r2_long |>
  filter(r2 < R2_FLOOR) |>
  group_by(analyte_id) |>
  summarise(v = min(r2), .groups = "drop") |>
  arrange(v) |>
  mutate(t = sprintf("%s %.3f", analyte_id, v))
p_r2 <- ggplot(d_r2_long, aes(r2, analyte_id)) +
  geom_line(aes(group = analyte_id), colour = "grey75", linewidth = 0.6) +
  geom_point(aes(colour = chain), size = 1.9, alpha = 0.9) +
  geom_text(
    data = r2_off,
    aes(label = "<"), hjust = 0, nudge_x = 0.00004, size = 2.3,
    colour = "grey45"
  ) +
  scale_colour_manual(values = chain_col, name = NULL) +
  scale_x_continuous(breaks = c(0.996, 0.997, 0.998, 0.999, 1.000)) +
  coord_cartesian(xlim = c(R2_FLOOR, 1.0004)) +
  labs(
    x = expression(Calibration ~ R^2),
```

```r
    y = NULL, title = "Calibration curve fit comparison",
    subtitle = paste0(
      "x-axis truncated at ",
      R2_FLOOR, "; off scale (<): ", paste(off_txt$t, collapse = ", ")
    )
  ) +
  theme(
    plot.subtitle = element_text(size = 6.5, colour = "grey30"),
    plot.margin = margin(4, 10, 4, 4), legend.position = "inside",
    legend.position.inside = c(0.015, 0.98),
    legend.justification.inside = c(0, 1),
    legend.background = element_rect(
      fill = alpha("white", 0.7),
      colour = NA
    ),
    legend.key.size = unit(3, "mm")
  )
p_r2
```

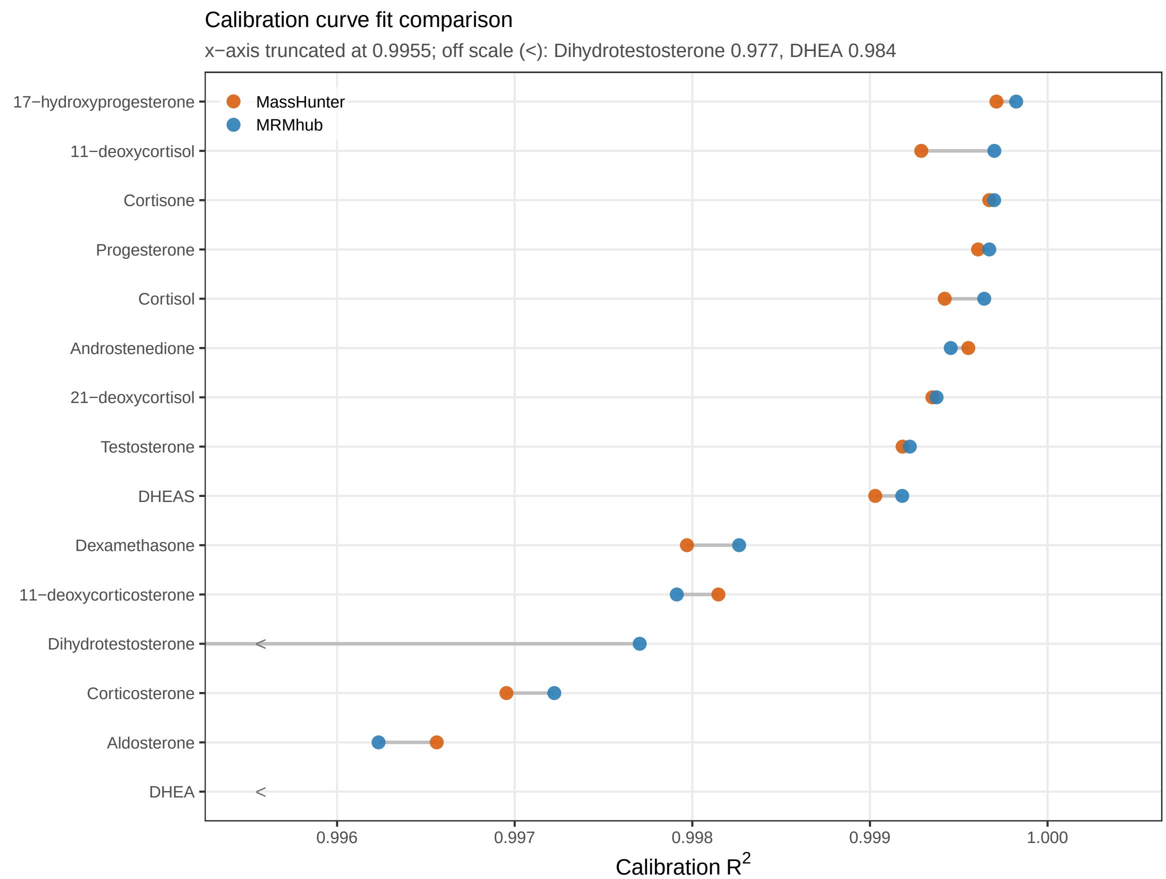

Figure 8: **Calibration-fit comparison.** R^2 per analyte, MassHunter's own fit vs the MRMhub full-pipeline fit. Panel A of Figure 12.

The same comparison as a table, with **three fits per analyte** (quadratic, `y = a·x² + b·x + c`, 1/x weighting): MassHunter's own fit, the MRMhub end-to-end fit (MRMhub-INTEGRATOR areas), and the MRMhub pipeline run on **MassHunter's own peak areas**. The gap between the two MRMhub rows is the peak-integration effect; the distance between *MRMhub (on MH areas)* and *MassHunter* isolates the post-processing (normalization, weighting, regression) alone.

```
library(gt)

# Measured LQC / HQC concentrations (nmol/L) per engine, one column each.
qc_wide <- function(df) {
  df |>
    filter(qc_type %in% c("LQC", "HQC")) |>
    pivot_longer(
      !c(analysis_id, qc_type),
      names_to = "analyte_id", values_to = "conc"
```

```
    ) |>
    select(analyte_id, qc_type, conc) |>
    pivot_wider(names_from = qc_type, values_from = conc)
}

tmp_mhconc <- tempfile(fileext = ".csv")
save_dataset_csv(
  mexp_mh,
  path = tmp_mhconc,
  variable = "conc",
  add_qctype = TRUE
)
#> v Concentration values for 30 analyses and 15 features have been exported to
 ↪  '/var/folders/3r/ywcsb3896zj4_0xlb_70yvlh0000gn/T//RtmptwQyra/file11b321c8d618.csv'.
conc_e2e <- qc_wide(d_conc_raw)
conc_onmh <- qc_wide(read_csv(tmp_mhconc, show_col_types = FALSE))
conc_mhun <- mh_data |>
  filter(analysis_id %in% c("QC_Low1_P1", "QC_High1_P1")) |>
  transmute(
    analyte_id = feature_id,
    qc_type = if_else(analysis_id == "QC_Low1_P1", "LQC", "HQC"),
    conc = conc_mh
  ) |>
  pivot_wider(names_from = qc_type, values_from = conc)

# Assemble one row per analyte for each of the three fits. mrmhub reports
# coefficients intercept-first (coef_a = x^0 … coef_c = x^2), so the two MRMhub
# blocks remap to MassHunter's descending order (a = x^2, b = x, c = intercept);
# MassHunter's export is already in that order.
mh_own <- read_csv(
  file.path(d4, "Dataset4_MassHunter-CalibrationFits.csv"),
  show_col_types = FALSE
) |>
  transmute(
    analyte_id,
    method = "MassHunter", a = coef_a, b = coef_b, c = coef_c, r2 = r2
  ) |>
  left_join(conc_mhun, by = "analyte_id")
# MRMhub pipeline fit on MassHunter's peak areas (isolates post-processing).
mrm_mh <- get_calibration_metrics(mexp_mh, include_qualifier = FALSE) |>
  transmute(
    analyte_id = feature_id,
    method = "MRMhub (on MH areas)", a = coef_c, b = coef_b, c = coef_a,
    r2 = r2
  ) |>
  left_join(conc_onmh, by = "analyte_id")
# MRMhub end-to-end fit on MRMhub-INTEGRATOR areas.
mrm_e2e <- get_calibration_metrics(mexp, include_qualifier = FALSE) |>
```

```r
  transmute(
    analyte_id = feature_id,
    method = "MRMhub (end-to-end)", a = coef_c, b = coef_b, c = coef_a,
    r2 = r2
  ) |>
  left_join(conc_e2e, by = "analyte_id")

cal_lv <- c("MassHunter", "MRMhub (on MH areas)", "MRMhub (end-to-end)")
fmt_conc <- function(x) {
  ifelse(
    is.na(x),
    "-",
    formatC(
      signif(x, 4),
      format = "fg", big.mark = ",", drop0trailing = TRUE
    )
  )
}
cal_coef_tbl <- bind_rows(mh_own, mrm_mh, mrm_e2e) |>
  mutate(method = factor(method, levels = cal_lv)) |>
  arrange(analyte_id, method) |>
  # Stripe by analyte group (3 rows each), not by row.
  mutate(.grp = match(analyte_id, unique(analyte_id))) |>
  transmute(
    .grp,
    Analyte = if_else(as.character(method) == cal_lv[1], analyte_id, ""),
    Method = as.character(method), `a (x²)` = sprintf("%.4g", a),
    `b (x)` = sprintf("%.4g", b), `c` = sprintf("%.4g", c),
    `R²` = sprintf("%.5f", r2), `LQC Concentration` = fmt_conc(LQC),
    `HQC Concentration` = fmt_conc(HQC)
  )

gt(cal_coef_tbl) |>
  cols_hide(".grp") |>
  cols_align(
    "right",
    columns = c(
      "a (x²)",
      "b (x)", "c", "R²", "LQC Concentration", "HQC Concentration"
    )
  ) |>
  # Zebra shading applied per analyte, so each analyte's three method rows
  # share one stripe and alternate against the next analyte. Both parities get
  # an explicit inline fill (odd = white) so Quarto's per-row table striping
  # can't leak through the unfilled groups.
  tab_style(
    cell_fill(color = "#FFFFFF"),
    cells_body(rows = .grp %% 2 == 1)
```

```r
  ) |>
  tab_style(
    cell_fill(color = "#F2F2F2"),
    cells_body(rows = .grp %% 2 == 0)
  ) |>
  # MRMhub end-to-end fit distinguished by blue text (see caption).
  tab_style(
    cell_text(color = "#2C7FB8"),
    cells_body(rows = Method == "MRMhub (end-to-end)")
  ) |>
  tab_options(table.font.size = px(12), data_row.padding = px(2))
```

5.2 Concentration equivalence on identical areas

Feeding MassHunter areas through the MRMhub pipeline reproduces MassHunter's end-to-end concentrations closely, indicating that the post-processing steps are effectively equivalent.

```r
tmp_conc <- tempfile(fileext = ".csv")
save_dataset_csv(
  mexp_mh,
  path = tmp_conc,
  variable = "conc",
  add_qctype = TRUE
)
#> v Concentration values for 30 analyses and 15 features have been exported to
↪  '/var/folders/3r/ywcsb3896zj4_0xlb_70yvlh0000gn/T//RtmptwQyra/file11b327f96ca74.csv'.
d_qonly <- read_csv(tmp_conc, show_col_types = FALSE) |>
  pivot_longer(
    !c(analysis_id, qc_type),
    names_to = "analyte_id", values_to = "conc_mrmpipe"
  ) |>
  left_join(
    mh_data |> select(analysis_id, analyte_id = feature_id, conc_mh),
    by = c("analysis_id", "analyte_id")
  ) |>
  filter(!is.na(conc_mrmpipe), !is.na(conc_mh)) |>
  filter(!qc_type %in% c("SBLK", "PBLK")) |>
  mutate(qc_type = factor(qc_type, levels = names(smp_cols)))
posq <- d_qonly |> filter(conc_mrmpipe > 0, conc_mh > 0)
q_r <- cor(log10(posq$conc_mrmpipe), log10(posq$conc_mh))
axq <- range(c(posq$conc_mrmpipe, posq$conc_mh))
p_qonly <- posq |>
  ggplot(aes(conc_mh, conc_mrmpipe, colour = qc_type)) +
  geom_abline(
    slope = 1,
```

Table 4: **Calibration fit per analyte.** Coefficients of $y = a \cdot x^2 + b \cdot x + c$, R^2, and measured LQC/HQC concentrations (nmol/L) for three fits — MassHunter's own, MRMhub on MassHunter areas, and MRMhub end-to-end (blue text). The two MRMhub rows differ only by peak integration; MRMhub-on-MH-areas vs MassHunter isolates post-processing.

Analyte	Method	a (x^2)	b (x)	c	R^2	LQC Concentration	HQC Con
11-deoxycorticosterone	MassHunter	0.01256	1.471	0.04589	0.99815	0.6529	
	MRMhub (on MH areas)	0.01256	1.471	0.0459	0.99815	0.6529	
	MRMhub (end-to-end)	0.01313	1.427	0.04811	0.99791	0.6723	
11-deoxycortisol	MassHunter	0.005944	0.9796	0.06228	0.99929	1.458	
	MRMhub (on MH areas)	0.005944	0.9796	0.06228	0.99929	1.458	
	MRMhub (end-to-end)	0.005598	1.022	0.0644	0.99970	1.451	
17-hydroxyprogesterone	MassHunter	0.001839	0.9685	0.03238	0.99971	1.488	
	MRMhub (on MH areas)	0.001839	0.9685	0.03238	0.99971	1.488	
	MRMhub (end-to-end)	0.001363	0.9671	0.03846	0.99982	1.513	
21-deoxycortisol	MassHunter	0.006042	0.7758	0.008574	0.99935	1.698	
	MRMhub (on MH areas)	0.006042	0.7758	0.008574	0.99935	1.697	
	MRMhub (end-to-end)	0.005988	0.7606	0.002312	0.99938	1.684	
Aldosterone	MassHunter	0.01507	0.5417	0.1124	0.99656	0.9199	
	MRMhub (on MH areas)	0.01507	0.5417	0.1124	0.99656	0.9199	
	MRMhub (end-to-end)	0.01406	0.518	0.08996	0.99623	0.9878	
Androstenedione	MassHunter	0.005103	0.6132	0.04322	0.99955	1.81	
	MRMhub (on MH areas)	0.005103	0.6132	0.04322	0.99955	1.81	
	MRMhub (end-to-end)	0.004824	0.6072	0.0427	0.99945	1.842	
Corticosterone	MassHunter	-6.449e-05	0.4008	-0.01636	0.99695	3.758	
	MRMhub (on MH areas)	-6.45e-05	0.4008	-0.01636	0.99695	3.757	
	MRMhub (end-to-end)	-6.664e-05	0.4109	-0.01616	0.99722	3.601	
Cortisol	MassHunter	1.227e-05	0.1237	-0.06246	0.99942	24.37	
	MRMhub (on MH areas)	1.227e-05	0.1237	-0.06246	0.99942	24.37	
	MRMhub (end-to-end)	1.526e-05	0.1212	-0.03548	0.99964	24.89	
Cortisone	MassHunter	-0.0003965	0.4315	-0.1158	0.99967	6.238	
	MRMhub (on MH areas)	-0.0003965	0.4315	-0.1158	0.99967	6.238	
	MRMhub (end-to-end)	-0.0003793	0.428	-0.09925	0.99970	6.269	
DHEA	MassHunter	-3.25e-05	0.03091	0.1772	0.98415	8.355	
	MRMhub (on MH areas)	-3.25e-05	0.03091	0.1772	0.98415	8.355	
	MRMhub (end-to-end)	-4.076e-05	0.03256	0.1832	0.98458	8.937	
DHEAS	MassHunter	-7.313e-09	0.003939	-0.0933	0.99903	524.2	
	MRMhub (on MH areas)	-7.313e-09	0.003939	-0.0933	0.99903	524.2	
	MRMhub (end-to-end)	-1.016e-08	0.003949	-0.09189	0.99918	521.2	
Dexamethasone	MassHunter	-1.546e-05	0.1036	-0.01331	0.99797	6.046	
	MRMhub (on MH areas)	-1.546e-05	0.1036	-0.01331	0.99797	6.046	
	MRMhub (end-to-end)	-1.714e-05	0.1036	-0.007976	0.99826	5.983	
Dihydrotestosterone	MassHunter	-0.03613	0.5642	-0.1789	0.97656	1.224	
	MRMhub (on MH areas)	-0.03613	0.5642	-0.1789	0.97656	1.224	
	MRMhub (end-to-end)	-0.01988	0.4645	-0.07173	0.99770	1.212	
Progesterone	MassHunter	-0.002326	0.9786	0.03361	0.99961	1.349	
	MRMhub (on MH areas)	-0.002326	0.9786	0.03361	0.99961	1.349	
	MRMhub (end-to-end)	-0.001442	0.9495	0.05202	0.99967	1.322	
Testosterone	MassHunter	0.003025	1.22	0.07571	0.99918	0.87	
	MRMhub (on MH areas)	0.003025	1.22	0.07571	0.99918	0.87	
	MRMhub (end-to-end)	0.002567	1.216	0.0762	0.99922	0.8558	

```r
    intercept = 0, linetype = "dashed", colour = "grey40"
  ) +
  geom_point(size = 1.1, alpha = 0.6) +
  annotate(
    "text",
    x = axq[1], y = axq[2], hjust = 0, vjust = 1, size = 2.4,
    label = sprintf("Pearson r = %.5f", q_r)
  ) +
  scale_x_log10(limits = axq) +
  scale_y_log10(limits = axq) +
  scale_colour_manual(
    values = smp_cols,
    drop = TRUE, name = "Sample type"
  ) +
  coord_equal() +
  labs(
    x = "MassHunter end-to-end (nmol/L)",
    y = "MRMhub-QUANT on MH areas (nmol/L)",
    title = "Concentration equivalence (same areas)"
  ) +
  theme(
    legend.position = "inside",
    legend.position.inside = c(0.98, 0.02),
    legend.justification.inside = c(1, 0),
    legend.background = element_rect(
      fill = alpha("white", 0.7),
      colour = NA
    ),
    legend.key.size = unit(2.8, "mm"),
    legend.title = element_text(size = 6.5),
    legend.text = element_text(size = 6)
  )
p_qonly
```

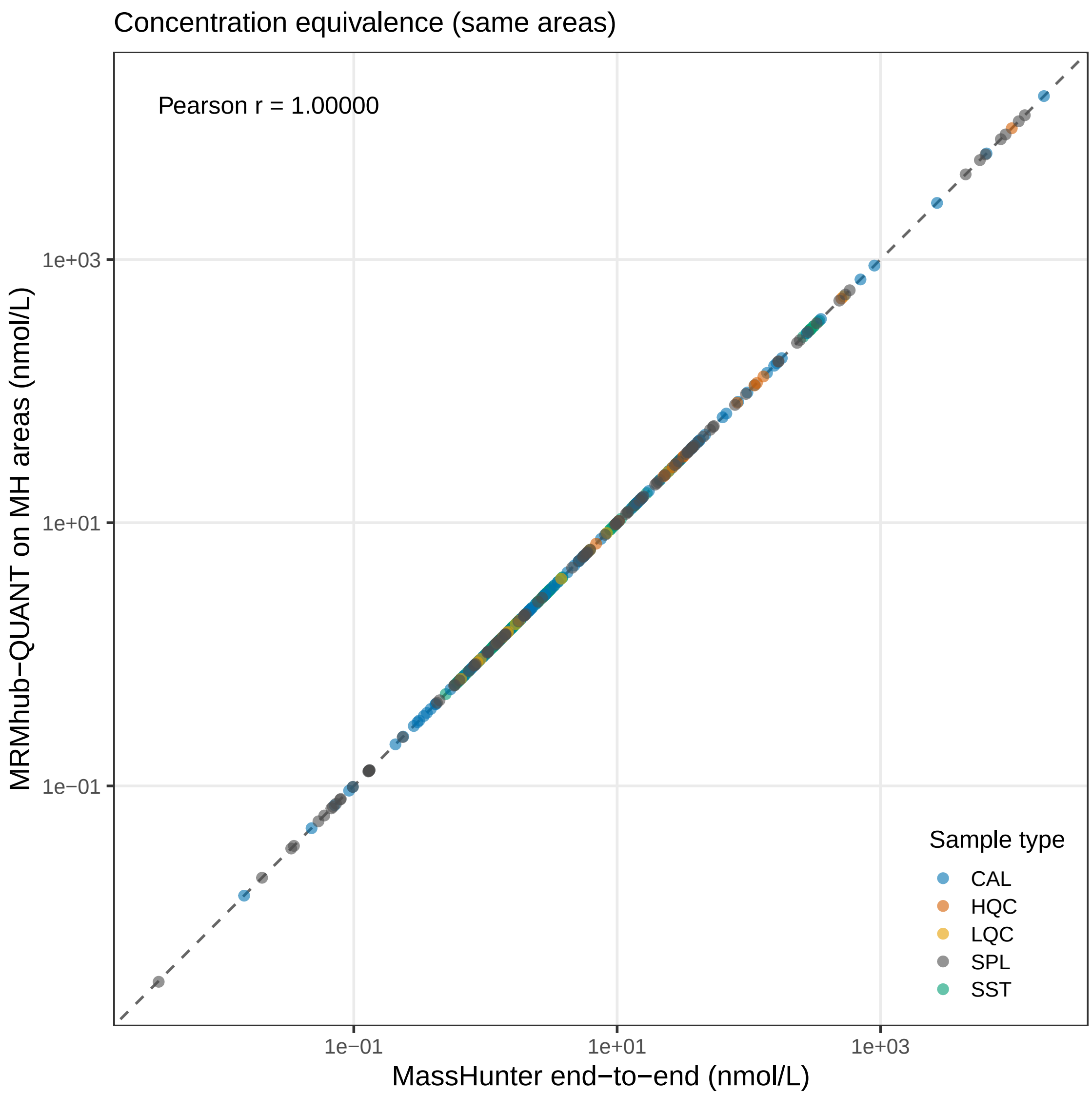

Figure 9: **Concentration equivalence on identical areas.** MassHunter areas run through the MRMhub pipeline vs MassHunter end-to-end, isolating post-processing from integration. Panel B of Figure 12.

5.3 Full end-to-end concentration agreement

Independent end-to-end concentrations (each method's own integration and calibration) for the LQC, HQC and SPL samples.

```
d_full <- d_conc_raw |>
  filter(qc_type %in% c("LQC", "HQC", "SPL")) |>
```

```r
  pivot_longer(
    !c("analysis_id", "qc_type"),
    names_to = "analyte_id", values_to = "conc"
  ) |>
  left_join(
    mh_data |> select(analysis_id, analyte_id = feature_id, conc_mh),
    by = c("analysis_id", "analyte_id")
  ) |>
  left_join(
    select(spl_area_excl, analysis_id, analyte_id, excl_mrm, excl_mh),
    by = c("analysis_id", "analyte_id")
  ) |>
  mutate(
    conc = if_else(coalesce(excl_mrm, FALSE), NA_real_, conc),
    conc_mh = if_else(coalesce(excl_mh, FALSE), NA_real_, conc_mh)
  ) |>
  select(-excl_mrm, -excl_mh) |>
  filter(!is.na(conc), !is.na(conc_mh), conc > 0, conc_mh > 0) |>
  mutate(qc_type = factor(qc_type, levels = names(smp_cols)))
f_r <- cor(log10(d_full$conc), log10(d_full$conc_mh))
axf <- range(c(d_full$conc, d_full$conc_mh))
p_full <- d_full |>
  ggplot(aes(conc_mh, conc, colour = qc_type)) +
  geom_abline(
    slope = 1,
    intercept = 0, linetype = "dashed", colour = "grey40"
  ) +
  geom_point(size = 1.3, alpha = 0.7) +
  annotate(
    "text",
    x = axf[1], y = axf[2], hjust = 0, vjust = 1, size = 2.4,
    label = sprintf("Pearson r = %.4f", f_r)
  ) +
  scale_x_log10(limits = axf) +
  scale_y_log10(limits = axf) +
  scale_colour_manual(
    values = smp_cols,
    drop = TRUE, name = "Sample type"
  ) +
  coord_equal() +
  labs(
    x = "MassHunter end-to-end (nmol/L)",
    y = "MRMhub end-to-end (nmol/L)",
    title = "Concentration agreement (full pipeline)"
  ) +
  theme(legend.position = "none")
p_full
```

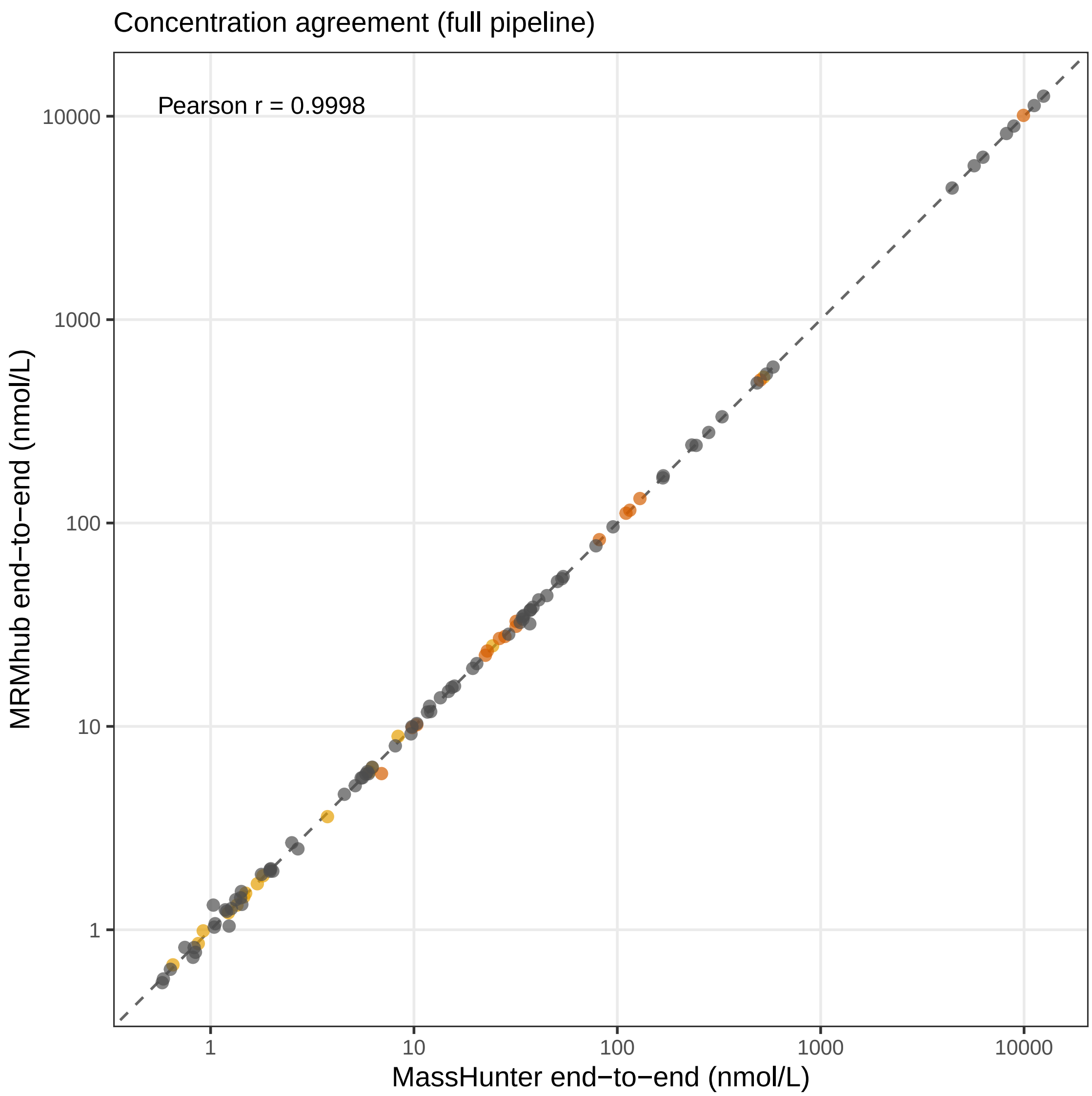

Figure 10: **Full end-to-end concentration agreement.** Each method's own integration and calibration, for the LQC, HQC and SPL samples. Panel C of Figure 12.

5.4 Accuracy against reference values (QC and EQA)

Per-analyte percent bias against the reference concentrations — the SKML EQA reference values and the nominal LQC/HQC target concentrations — one point per sample per method (dashed lines: ±15 %; LQC and HQC are grouped to the left of the EQA samples). EQA points whose quantifier area falls below the lowest calibrator (Cal A, i.e. below the limit of quantification) are drawn as open symbols rather than excluded. The reference concentrations are read from the metadata workbook's `QC Concentrations` sheet (imported above with the rest of the metadata), keyed by sample and analyte. DHEA and DHEAS are excluded (large biases that would compress the other panels).

```
# Reference (target) concentrations come from the metadata's `QC Concentrations`
# sheet (imported above), keyed by `sample_id` + `analyte_id`, and surface via the
# `annot_qcconcentrations` slot (all in nmol/L). Both the SKML EQA reference values
# and the nominal LQC/HQC targets are kept, so accuracy is assessed for all three.
acc_target <- mexp@annot_qcconcentrations |>
  filter(str_starts(sample_id, "SKML") | sample_id %in% c("LQC", "HQC")) |>
  transmute(sample_id, analyte_id, target = concentration)

# analysis_id -> sample_id map (groups the `…b` replicates onto one EQA sample).
sid_map <- get_analyticaldata(mexp, annotated = TRUE) |>
  distinct(analysis_id, sample_id)

d_acc <- d_conc_raw |>
  filter(qc_type %in% c("LQC", "HQC", "SPL")) |>
  pivot_longer(
    !c("analysis_id", "qc_type"),
    names_to = "analyte_id", values_to = "conc_mrm"
  ) |>
  left_join(
    mh_data |> select(analysis_id, analyte_id = feature_id, conc_mh),
    by = c("analysis_id", "analyte_id")
  ) |>
  left_join(
    select(spl_area_excl, analysis_id, analyte_id, excl_mrm, excl_mh),
    by = c("analysis_id", "analyte_id")
  ) |>
  left_join(sid_map, by = "analysis_id") |>
  inner_join(acc_target, by = c("sample_id", "analyte_id")) |>
  # Compact axis label: SKML EQA -> 2026.1A; LQC/HQC keep their name.
  mutate(
    sample_lbl = str_replace(
      str_remove(sample_id, "^SKML"),
      "^([0-9]{4})_", "\\1."
    )
  ) |>
  # Below-Cal-A (below-LOQ) values are kept rather than masked; a per-method flag
  # (`blo_*`) lets the panel draw them as open symbols. LQC/HQC carry no exclusion
  # flag (never below-Cal-A here), so they count as quantifiable.
  transmute(
    analyte_id,
    sample_lbl, analysis_id, target, conc_mrm, conc_mh,
    bias_mrmhub = (conc_mrm - target) / target * 100,
    bias_masshunter = (conc_mh - target) / target * 100,
    blo_mrmhub = coalesce(excl_mrm, FALSE),
    blo_masshunter = coalesce(excl_mh, FALSE)
```

```r
  ) |>
  filter(!(is.na(conc_mrm) & is.na(conc_mh)))

# x-axis order: LQC, HQC first, then the EQA samples (sorted by label).
eqa_lbls <- sort(setdiff(unique(d_acc$sample_lbl), c("LQC", "HQC")))
x_levels <- c("LQC", "HQC", eqa_lbls)

p_eqa <- d_acc |>
  filter(!analyte_id %in% c("DHEA", "DHEAS")) |>
  # Reshape bias and the below-LOQ flag together (one row per method).
  pivot_longer(
    c(bias_mrmhub, bias_masshunter, blo_mrmhub, blo_masshunter),
    names_to = c(".value", "method"),
    names_pattern = "(bias|blo)_(mrmhub|masshunter)"
  ) |>
  filter(!is.na(bias)) |>
  mutate(
    method = recode(method, mrmhub = "MRMhub", masshunter = "MassHunter"),
    method = factor(method, levels = c("MRMhub", "MassHunter")),
    sample_lbl = factor(sample_lbl, levels = x_levels),
    loq = factor(
      if_else(blo, "Below LOQ (< Cal A)", "Quantifiable (>= Cal A)"),
      levels = c("Quantifiable (>= Cal A)", "Below LOQ (< Cal A)")
    )
  ) |>
  ggplot(aes(
    sample_lbl, bias,
    colour = method, shape = loq, group = method
  )) +
  geom_hline(yintercept = 0, colour = "grey40") +
  geom_hline(
    yintercept = c(-15, 15),
    linetype = "dashed", colour = "grey75"
  ) +
  # Divider between the LQC/HQC group (left) and the EQA samples (right).
  geom_vline(xintercept = 2.5, colour = "grey85", linewidth = 0.3) +
  geom_point(
    size = 1.5,
    alpha = 0.85, position = position_dodge(width = 0.4)
  ) +
  scale_colour_manual(values = col_method, name = "Method") +
  # Open symbols flag EQA values below the lowest calibrator (below LOQ).
  scale_shape_manual(
    values = c("Quantifiable (>= Cal A)" = 16, "Below LOQ (< Cal A)" = 1),
    name = NULL
  ) +
  facet_wrap(vars(analyte_id), scales = "free_y", ncol = 3) +
  guides(
```

```r
    colour = guide_legend(override.aes = list(shape = 16)),
    shape = guide_legend(override.aes = list(colour = "grey30"))
  ) +
  labs(
    x = NULL,
    y = "Bias vs reference (%)",
    title = "Accuracy - bias per analyte (QC & EQA; excl. DHEA, DHEAS)"
  ) +
  theme(
    aspect.ratio = 1,
    panel.grid.minor = element_blank(),
    strip.text = element_text(size = 6),
    axis.text.x = element_text(angle = 45, hjust = 1, size = 5),
    axis.text.y = element_text(size = 5.2)
  )
p_eqa
```

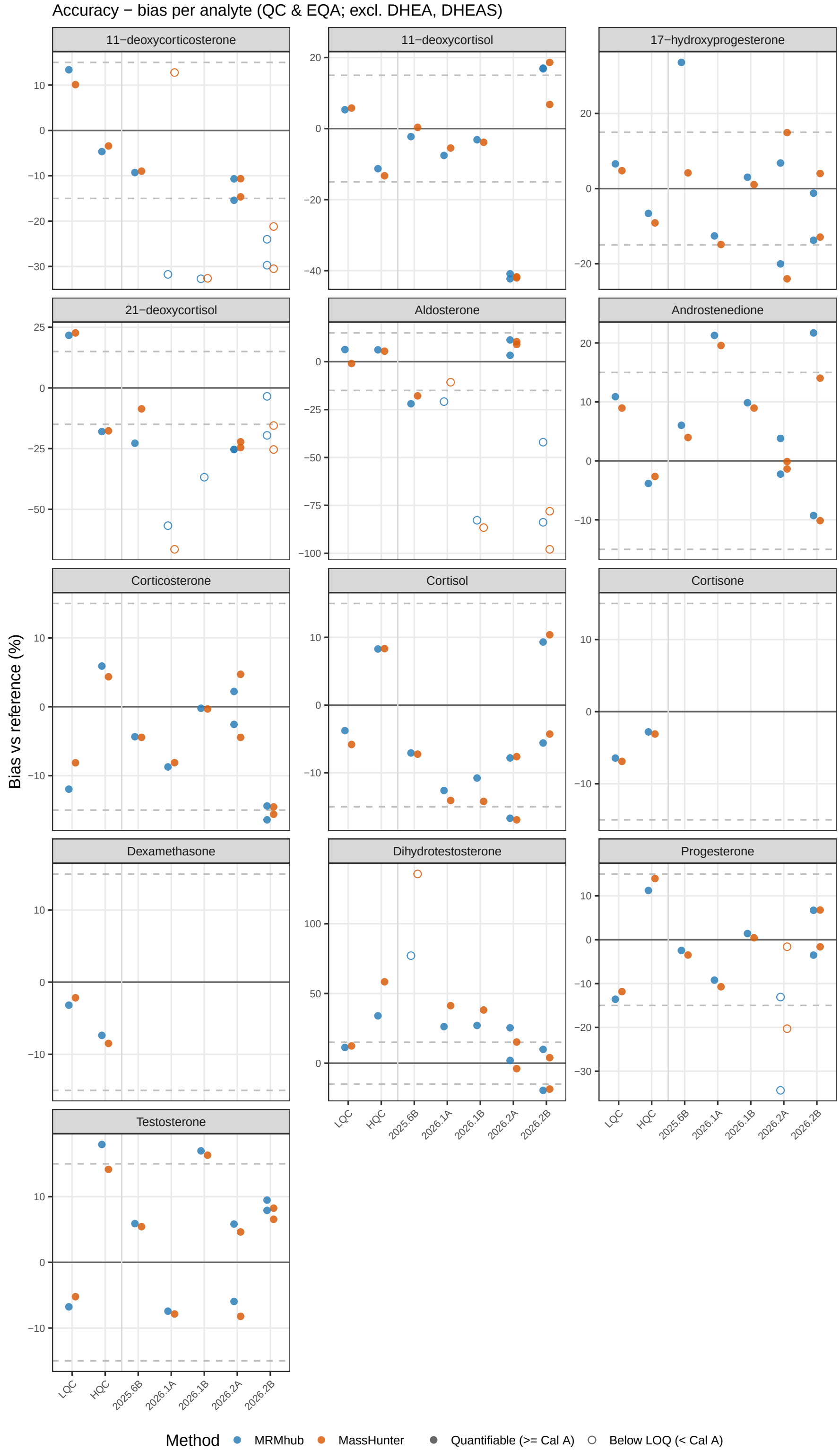

Figure 11: **Accuracy against reference values.** Per-analyte percent bias against the reference concentrations — SKML EQA values plus the nominal LQC/HQC targets (±15 % dashed) — MRMhub vs MassHunter, one point per sample per method, with LQC/HQC grouped to the left of the EQA samples; open symbols mark EQA values below the lowest calibrator (Cal A, below LOQ) and DHEA and DHEAS are excluded. Panel D of Figure 12.

5.5 Figure 2 — manuscript figure

The four panels above are composed into the publication figure (**Figure 2**).

```
# Middle row (B, C) uses coord_equal(); give it enough height that the two square
# panels grow to fill the full width, so their L/R borders align with A and D.
# The bias panel is a square, 3-column grid as a standalone figure; inside the
# full-width composite it fills better as a wide 4-column grid without the forced
# square aspect, so override those two settings for Figure 2 only.
p_eqa_fig2 <- p_eqa +
  facet_wrap(vars(analyte_id), scales = "free_y", ncol = 4) +
  theme(aspect.ratio = NULL)
fig2 <- p_r2 /
  (p_qonly + p_full) /
  p_eqa_fig2 +
  plot_layout(heights = c(0.68, 1.3, 1.2)) +
  plot_annotation(tag_levels = "A")

save_fig(fig2, "fig2_quantification_comparison", height = 236)
fig2
```

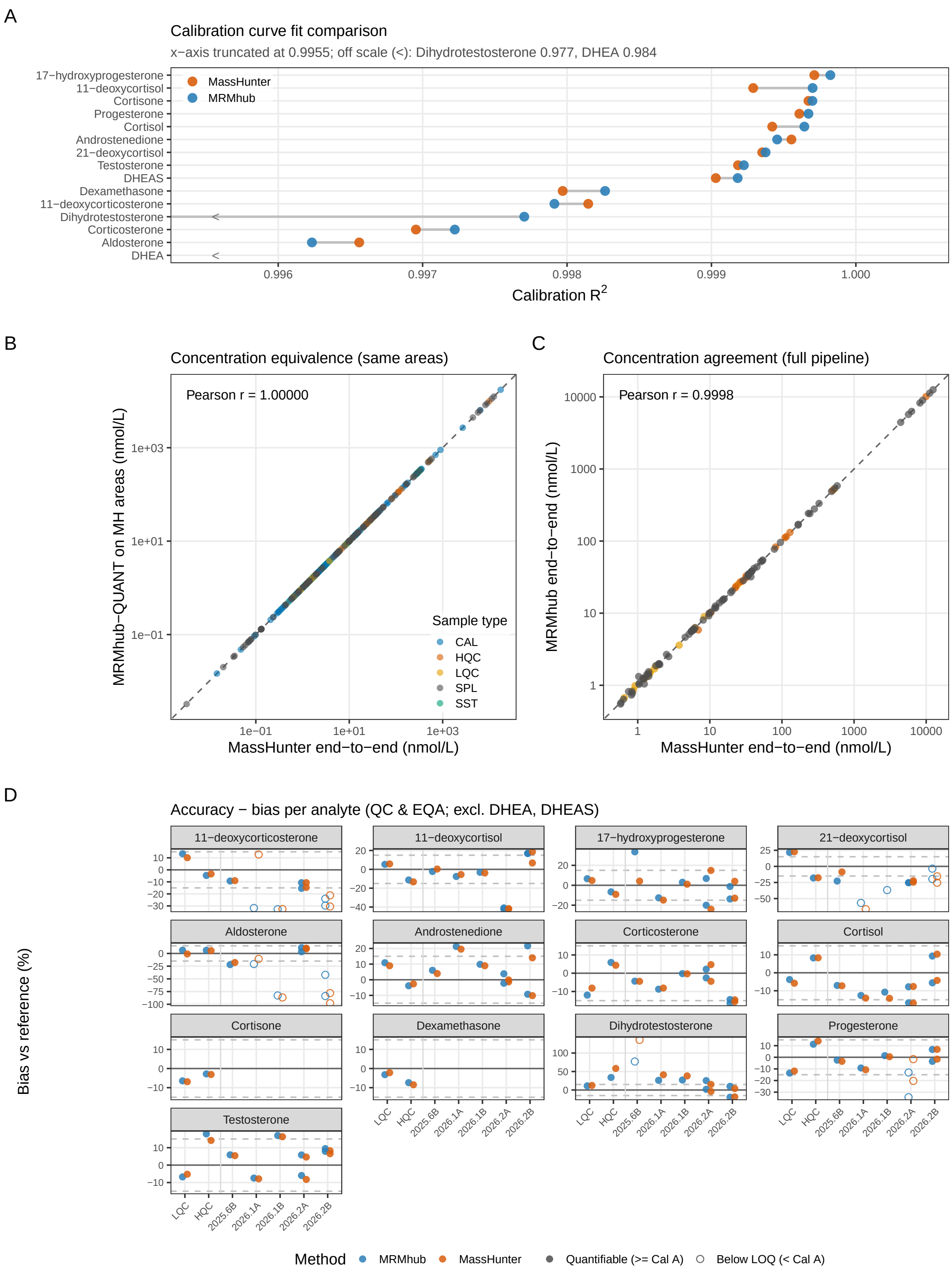

Figure 12: **MRMhub vs MassHunter — quantification.** Composite of the four panels above; reproduced as the manuscript's **Figure 2**. **(A)** Calibration-fit comparison, **(B)** concentration equivalence on identical areas, **(C)** full end-to-end agreement, **(D)** QC and EQA accuracy. Exported to `output/fig2_quantification_comparison.{pdf,png}` (180 mm).

6 Summary

Across both pipeline stages, MRMhub-QUANT and Agilent MassHunter agree closely on this steroid panel. At the **integration** stage, peak areas track the 1:1 line over four orders of magnitude with comparable replicate precision, and the few larger real-sample differences trace to legitimate integration-boundary choices on elevated or co-eluting baselines rather than to a systematic bias (Figure 7). At the **quantification** stage, feeding MassHunter's own areas through the MRMhub pipeline reproduces MassHunter's concentrations, isolating the post-processing as effectively equivalent; the two independent end-to-end pipelines then show comparable accuracy against the nominal LQC/HQC targets and the SKML EQA reference values (Figure 12). Any residual differences are attributable to peak integration rather than to the QUANT post-processing.

References

IBL International GmbH. 2022. *Steroid Panel LC-MS (Cat. 30191875)*. Instructions for Use, rev. 2022-06, IBL International GmbH (a Tecan Group company), Hamburg, Germany.

Jansen, Rob, Nuthar Jassam, Annette Thomas, et al. 2014. "A Category 1 EQA Scheme for Comparison of Laboratory Performance and Method Performance: An International Pilot Study in the Framework of the Calibration 2000 Project." *Clinica Chimica Acta* 432 (May): 90–98. https://doi.org/10.1016/j.cca.2013.11.003.